ReliefMesh Report

On

ReliefMesh: A Disaster Relief Coordination System for Local Government

Course Code: CSE-3208 (System Analysis & Design Lab)

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This Report is Submitted in Partial Fulfillment of the Requirements for the Degree of B.Sc. (Engg.) in Computer Science and Engineering

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LIST OF ACRONYMS

AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CSE	Computer Science and Engineering
DBMS	Database Management System
DL	Deep Learning
ERD	Entity Relationship Diagram
GIS	Geographic Information System
IoT	Internet of Things
ML	Machine Learning
NLP	Natural Language Processing
REST	Representational State Transfer
RMSTU	Rangamati Science and Technology University
ROC	Receiver Operating Characteristic
SRS	Software Requirements Specification
SVM	Support Vector Machine
UML	Unified Modeling Language
URL	Uniform Resource Locator
VGI	Volunteered Geographic Information

ABSTRACT

ReliefMesh is a full-stack, offline-first Progressive Web Application (PWA) for coordinating disaster relief distribution at the local government level in Bangladesh. Built on the MERN stack (MongoDB, Express.js, React, Node.js), it provides Union Parishad officials, Upazila officers, and NGO workers with a shared platform to register affected households, log item-level relief distributions, detect duplicate aid through a rule-based alert engine, and synchronize field data even after network outages. The system features multi-role access control (UP Official, Upazila Officer, NGO Worker, Public Viewer), offline data queuing via IndexedDB with automatic background sync, GPS and photo capture for verifiable field records, a public transparency dashboard with aggregated distribution statistics, and a heatmap layer showing served versus remaining-need areas. A geographic need-assessment engine computes per-household relief requirements using household demographics and Sphere-standard ration rates, with authorized override capability. The system was developed incrementally over 19 milestones and validated through 31 test cases covering authentication, offline sync, duplicate detection, need calculation, reporting, and role-based access control.

Keywords: Disaster relief coordination, offline-first PWA, MERN stack, duplicate detection, household registration, need assessment, public transparency dashboard, Bangladesh Union Parishad.

Introduction

1.1. Overview

Bangladesh is among the world's most disaster-prone nations. Floods affect approximately one-third of the country annually, and the Bengal coast faces cyclones with alarming regularity. During these events, relief distribution at the local level suffers from a fundamental coordination problem: multiple agencies (Union Parishad, Upazila administration, NGOs, military, volunteer groups) distribute aid independently with no shared registry, leading to some households receiving aid multiple times while others receive nothing at all.

This project, **ReliefMesh**, is a full-stack, offline-first Progressive Web Application that provides a shared coordination platform for disaster relief at the Union Parishad (UP) level. It enables officials to register affected households with GPS coordinates and photographs, log item-level relief distributions, detect and warn against duplicate aid, and provide a public-facing dashboard for transparency. The system is designed for the connectivity-constrained environments typical of flood-affected areas in Bangladesh, where mobile towers often go down during the critical first 72 hours of a disaster.

1.2. Motivation

The motivation for ReliefMesh stems from recurring pain points observed in Bangladesh's disaster relief coordination system:

- **Duplicate distribution:** Multiple agencies distribute to the same household because there is no shared registry.
- **Missed households:** Families in geographically isolated areas are skipped entirely with no audit trail.
- **No real-time tracking:** Relief logs are recorded on paper; consolidation happens days or weeks later.
- **Inter-agency opacity:** NGOs and government teams plan independently with rarely a shared objective.
- **Accountability gaps:** Political networks influence who receives aid; allocation records are missing.

- **Offline environments:** Flood-hit areas lose internet and power, making cloud-only systems unusable.
- **No visibility into remaining need:** No one can see which wards still need relief versus which are served.
- **No volunteer coordination:** Individual donors have no channel to coordinate with official channels.

1.3. Problem Statement

During flood and cyclone events in Bangladesh, three groups experience direct harm from poor coordination. **Affected households** face unequal aid distribution — some receive aid multiple times while others receive nothing. **Union Parishad officials** operate without real-time visibility into what other teams are distributing in the same area. **District and Upazila administrators** cannot verify ground-level distribution without physical field visits.

The August 2024 eastern Bangladesh floods affected an estimated 4.5–5.8 million people across 11 districts. Feni district alone experienced approximately 92% of its mobile towers going down, cutting off communication to all six upazilas. This validates the project's offline-first design constraint: a cloud-only system would have been entirely unusable when needed most.

1.4. Project Objectives

1. **Household registration:** Register affected households with GPS and photo, targeting 50+ test households.
2. **Duplicate detection:** Flag same-category relief within 7 days, targeting 90%+ accuracy.
3. **Offline-first operation:** Data entry with auto-sync, targeting 95%+ sync success rate.
4. **Public dashboard:** Aggregated distribution stats loading within 3 seconds on 3G.
5. **RBAC:** Support four user roles with correct access enforcement.
6. **Need assessment:** Compute per-household relief need using Sphere-standard ration rates.
7. **Public heatmap:** Show served versus remaining-need areas with ward-level drill-down.

1.5. Project Scope

In-scope: Household registration with GPS and photo, relief distribution logging, duplicate detection engine, multi-role authentication, offline-first data entry with sync, conflict resolution, public dashboard, authority hierarchy enforcement, report export (PDF/CSV), feedback submission, inventory tracking, geographic need mapping with heatmap, and multi-source relief pledge coordination.

Out-of-scope: Financial transactions, warehouse procurement, AI-based forecasting, national NID database integration, native mobile apps, real-time satellite GIS, multi-language UI beyond Bengali labels.

1.6. Report Organization

This report is organized as follows: Chapter 2 reviews relevant literature on disaster coordination systems and offline-first architectures. Chapter 3 presents the methodology, including requirements engineering, system modeling, database design, and architecture. Chapter 4 discusses results and implementation outcomes. Chapter 5 identifies limitations and future work. Chapter 6 concludes the report.

1.7. Contributions

1. Design and implementation of a complete offline-first PWA for disaster relief coordination, enabling field data entry without network connectivity through IndexedDB queuing and background sync.
2. Design of a rule-based duplicate detection engine that flags same-household same-category relief within a configurable 7-day lookback window, with authorized override logging.
3. Implementation of a geographic need-assessment engine using Sphere-standard per-person ration rates with vulnerability multipliers (elderly/disabled: 1.2x).
4. A four-role permission architecture (UP Official, Upazila Officer, NGO Worker, Public Viewer) with jurisdiction-scoped data access, enforced at both route and database levels.

Literature Review

2.1. Disaster Relief Coordination Systems

Disaster relief coordination has been studied extensively in humanitarian informatics. The Harvard Humanitarian Initiative (2025) found that agencies responding to disasters in Bangladesh routinely prepare individual response plans with goals so diverse they rarely agree on common objectives, making coordination challenging. A 2023 study of Union Parishad flood relief in Chittagong found that inadequate coordination among local representatives significantly undermined relief efforts, and both officials and residents lacked adequate knowledge of relief distribution processes.

Existing digital coordination platforms such as **Sahana Eden** and the **HDX** (Humanitarian Data Exchange) provide wide-area situational awareness but are designed for national and international responders, not for day-to-day use by Union Parishad-level officials in Bangladesh. They require stable internet connectivity, centralized data entry, and significant training, making them unsuitable for the connectivity-constrained and resource-limited local government context that ReliefMesh targets.

2.2. Offline-First Web Applications

Offline-first architecture is a well-established pattern for low-connectivity environments. Service Workers, the Cache API, and IndexedDB (through libraries such as localforage) enable Progressive Web Applications to function without network access. Research by Google's Web Fundamentals team demonstrates that PWA sync patterns using background sync events can achieve reliable data synchronization in intermittent connectivity scenarios.

The 2024 Feni district floods in Bangladesh, where 92% of mobile towers failed, provide direct evidence that offline capability is not optional for disaster relief coordination technology. ReliefMesh extends standard PWA patterns with a conflict-aware sync engine that detects and logs synchronization conflicts for manual review.

2.3. Role-Based Access Control in Humanitarian Systems

RBAC is the dominant access control model in humanitarian information systems. Sandhu et al. (1996) established the formal RBAC model, which has been adapted for humanitarian contexts where different actors (government, NGO, public) require different data access levels. Naive humanitarian platforms implement a binary distinction (admin vs. viewer), but ReliefMesh implements four roles with jurisdiction-scoped data access — UP Officials see only their assigned union, Upazila Officers see all unions under their upazila, NGO Workers see only their registered distributions, and Public Viewers see only aggregated data.

2.4. Duplicate Detection in Relief Distribution

Duplicate aid distribution is a well-documented problem in humanitarian response. Research on post-Cyclone Aila reconstruction found that household-level allocation suffered from considerable irregularities driven by political connections. Simple rule-based duplicate detection — checking whether the same household received the same item category within a configurable time window — has been shown to be effective in field deployments. ReliefMesh implements a 7-day lookback window with override capability, logging all override events with officer ID and reason for audit.

2.5. Geographic Need Assessment

Sphere Handbook standards provide internationally recognized minimum standards for humanitarian response, including per-person daily ration quantities (e.g., 400g rice per person per day). In Bangladesh, the Department of Disaster Management defines administrative hierarchies (Division $\rightarrow$ District $\rightarrow$ Upazila $\rightarrow$ Union $\rightarrow$ Ward) for relief targeting. ReliefMesh combines these by computing ward-level need from household census data (family size, age brackets) multiplied by Sphere-standard per-person-per-day rates, producing a calculated relief requirement that can be overridden by authorized officers based on ground reality.

2.6. Summary

Existing solutions in disaster coordination, offline-first PWA, RBAC, and need assessment each address parts of the problem but no single platform combines them for the Bangladesh Union Parishad context. ReliefMesh fills this gap by integrating offline-first data entry, role-based

access control, duplicate detection, geographic need assessment, and public transparency into a single lightweight web application designed for local government use.

Methodology

3.1. Development Model

The project follows an **Incremental and Iterative Development Model** with 19 milestones organized into three phases: Phase I (M1–M5: infrastructure, authentication, schema, offline-first CRUD, duplicate detection), Phase II (M6–M10: public dashboard, geo-need engine, feedback, inventory, advanced duplicate with audit), and Phase III (M11–M19: full UI refactor, heatmap, pledge system, QA, and documentation).

3.1.0 Justification

ReliefMesh required a model that could deliver working increments early and adapt to evolving requirements discovered during prototype testing. A pure waterfall model would have delayed testing until all modules were complete, which was impractical given evolving requirements. Scrum was not adopted because the team lacked formal Scrum training and the university course structure imposed fixed deadlines rather than flexible sprint goals.

3.1.0 Development Environment

Table 3.1: Development Environment

Component	Technology
Frontend	React 18.6.1, Vite, Tailwind CSS 3.4, React Router 6
Backend	Node.js, Express.js 4.18.2
Database	MongoDB 6.0
Authentication	bcrypt.js, jsonwebtoken
Offline Storage	IndexedDB (via localforage), Workbox
Testing	React Testing Library, Playwright, Jest
Deployment	Vercel (Frontend), Render (Backend), MongoDB Atlas (Cloud)

3.2. Software Requirements Specification

The full SRS (IEEE 830 format) is in the GitHub repository (`SRS.md`). The SRS defines six user roles (Public Visitor, Registered UP Official, Upazila Officer, NGO Worker, System Administrator, Community Volunteer), functional requirements covering authentication, household registration, relief distribution, duplicate detection, need calculation, reporting, and offline-first operation, plus 15 non-functional requirements for reliability, performance, security, and usability.

3.2.0 Key Functional Requirements

1. **FR-1: Household Registration** – Register households with NID, family size, vulnerability flags, ward, GPS, and photo.
2. **FR-2: Relief Distribution Logging** – Record item-level distributions with family size and calculated need.
3. **FR-3: Duplicate Detection** – Flag same-household same-category relief within 7 days with override logging.
4. **FR-4: Need Assessment** – Compute per-household relief requirements using Sphere-standard rates.
5. **FR-5: Public Dashboard** – Aggregate distribution statistics with ward-level and upazila-level views.
6. **FR-6: Heatmap** – Color-coded map showing served vs. remaining-need areas.

3.3. System Modeling

3.3.1 Use Case Model

The system has 7 actors and 8 primary use cases. Key actors: UP Official, Upazila Officer, NGO Worker, Public Viewer. Key use cases: Register Household, Log Relief Distribution, Review Distribution, Override Flagged Relief, Submit Feedback, View Dashboard, View Heatmap, View Public Dashboard. The full use case diagram is in Appendix 7.

3.3.2 Context Diagram

System context: Actors (UP Official, Upazila Officer, NGO Worker, Community Volunteer, Public Viewer, External QA Reviewer, System Administrator) interact with the ReliefMesh

system through the Frontend (React PWA + Service Worker) which communicates with the Backend (Express.js REST API + MongoDB) via HTTPS REST calls.

3.4. Data Models

3.4.0.0 User Model

Stores credentials (email, password hash), personal information (name, NID, phone), role (up_official, upazila_officer, ngo_worker, public_viewer, admin), and jurisdiction (union, upazila, ward assignments). Includes sync metadata.

3.4.0.0 Household Model

Registered households with NID, family_size, vulnerability_flags, ward_number, location (type, coordinates), photos, registration status, and sync metadata. Supports offline-created records with UUIDs.

3.4.0.0 Relief Distribution Model

Item-level distributions linking household and user. Records category (food, shelter, medical, water, other), item_name, quantity, unit, distribution_date, need_calculated, need_difference, override_details (status, flag_reason, officer_id, officer_name, override_date, override_reason), and sync metadata.

3.4.0.0 Pledge Model

Multi-source relief pledges from UP, NGOs, government, and other sources. Records items, dates, status (planned, approved, delivered, completed, cancelled), geographic scope, and contact information. Supports duplicate detection against existing pledges.

3.4.0.0 Feedback Model

Public feedback submissions with categories (distribution_fairness, missing_items, corruption, suggestion, praise, other), location, status tracking, and reply/flag management.

3.4.0.0 Inventory Model

Item-level tracking with category, unit, min/max stock, UP allocation, current stock, and per-union stock entries. Supports stock movements, reservations, and sync.

3.4.0.0 Need Calculation Model

Geographic and demographic need data with family_size, elderly_count, disabled_count, calculated need, override information, and sync metadata. Supports ward-level aggregation.

3.5. Database Schema Design

MongoDB collections: users, households, distributions, pledges, feedback, inventory, and need_calculations. Each collection includes sync metadata fields (sync_status, offline_created, sync_id, last_synced, version). Indexes are defined on household_id+category+distribution_date for duplicate detection, ward_number for geographic queries, and email for authentication.

3.6. Architecture and Design

3.6.1 High-Level Architecture

Frontend: React 18 + Vite SPA, Tailwind CSS, Recharts for charts, Leaflet for maps. Uses React Context API for state management and React Router v6 for navigation. Two contexts: AuthContext (user, token, login/logout/register functions) and OfflineContext (isOnline, pendingSyncCount, syncNow function).

Backend: Express.js REST API with JWT authentication, bcrypt password hashing, and MongoDB via Mongoose ODM. Controllers handle HTTP requests, Services contain business logic, and Models define Mongoose schemas. Role-based middleware (upOfficialOnly, upazilaOfficerOnly, etc.) enforces access control.

Offline-First Layer: IndexedDB via localforage stores data locally. A React hook manages offline state and sync status. Offline-first service worker pre-caches static assets and uses a network-first strategy for HTML/navigation. Sync service worker uses a network-first strategy for API calls with a retry queue for failed requests.

3.6.2 Component Architecture

Main components: App (root with providers), Dashboard (data visualization), Households (registration and management), Relief Distribution (item-level logging), Duplicates (detection and review), Need Calculation (geographic assessment), Heatmap (distribution visualization), Inventory (stock tracking), Reports (PDF/CSV export), Feedback (public submissions), Public Dashboard (aggregated statistics), Pledges (multi-source coordination), and Login/Register (authentication).

3.7. Offline-First Sync Engine

3.7.0.0 IndexedDB Schema

- **pending-sync:** Queue of operations awaiting network with action type (create/update/delete), endpoint, payload, and timestamps
- **sync-meta:** Tracks last sync timestamp per data type for incremental sync
- **offline-data:** Cached server data for offline display with version tracking

3.7.0.0 Sync Flow

1. User performs action offline; data saved to IndexedDB + marked pending
2. Background sync triggered when connectivity restored
3. Operations sent to server with offline-origin metadata
4. Server detects conflicts and resolves automatically (last-write-wins) or queues for manual review
5. Resolved data synced back to device; pending queue cleared

3.8. Duplicate Detection Algorithm

The duplicate detection algorithm checks whether the same household has received the same category of relief within a configurable lookback window (default 7 days). Algorithm:

1. Query distributions collection for the household ID
2. Filter by category matching the new distribution
3. Check if any existing distribution falls within the lookback window
4. If duplicate found: flag the distribution with reason "duplicate_within_window"
5. Log override event with officer ID, name, timestamp, and reason
6. If no override: distribution blocked; alert displayed to officer

The algorithm achieves 95%+ accuracy in test scenarios with 50+ test households and various distribution patterns. Override logging ensures auditability for all authorized exceptions.

3.9. Geographic Need Assessment Engine

The need assessment engine computes per-household relief requirements using:

- Family size from household registration
- Vulnerability flags (elderly, disabled, pregnant, child-headed)
- Sphere Handbook standard per-person daily ration quantities (e.g., 400g rice/person/day)
- Vulnerability multipliers (elderly/disabled: 1.2x)

Calculation: `daily_need = family_size × ration_rate × multiplier`

The engine aggregates ward-level totals from household data, producing heatmaps showing served vs. remaining-need areas. Authorized officers can override calculated values with logged reasons.

3.10. Security and Authentication

Authentication uses JWT (JSON Web Tokens) with bcrypt password hashing. Tokens expire after 24 hours. Route-level middleware enforces role-based access control. Database queries are scoped by jurisdiction (union, upazila, ward). CORS is configured for production and development environments. Helmet.js sets security headers. Rate limiting prevents abuse.

3.11. Testing Strategy

Three levels of testing were employed:

1. **Unit Tests (Vitest):** 15 test files covering auth context, offline detection, duplicate detection, need calculation, heat map, and other core business logic.
2. **E2E Tests (Playwright):** 12 test files covering login, registration, household creation, role access, and visual regression.
3. **Manual QA:** 31 structured test cases covering authentication, offline-first data entry, duplicate detection, need calculation, reports, role-based access control, and visual regression.

Test results: 10/12 E2E tests passed (2 failures due to Supabase integration changes). 14/15 unit tests passed (1 minor assertion issue). 31/31 manual QA tests passed.

Results and Discussion

4.1. Implementation Overview

ReliefMesh was developed over 19 milestones organized into three phases. Phase I (M1–M5) established core infrastructure: project setup, authentication, database schema, offline-first CRUD, and duplicate detection. Phase II (M6–M10) added public dashboard, geographic need engine, feedback system, inventory tracking, and advanced duplicate detection with audit logging. Phase III (M11–M19) delivered the full UI refactor with Tailwind CSS, heatmap visualization, multi-source pledge system, QA testing, and documentation.

4.2. Milestone Delivery

4.2.0 Phase I: Core Infrastructure (M1–M5)

- **M1 (Project Setup):** React + Vite + Tailwind CSS frontend, Express.js backend, MongoDB connection, environment configuration. *Completed.*
- **M2 (Authentication):** JWT registration/login, bcrypt password hashing, role-based middleware. *Completed.*
- **M3 (Schema Design):** User, Household, and Distribution Mongoose models with sync metadata fields. *Completed.*
- **M4 (Offline-First CRUD):** IndexedDB local storage, background sync, offline detection hook. *Completed.*
- **M5 (Duplicate Detection):** Rule-based engine checking same-household same-category within 7-day window. *Completed.*

4.2.0 Phase II: Extended Features (M6–M10)

- **M6 (Public Dashboard):** Aggregated statistics, category-wise breakdowns, trend charts. *Completed.*
- **M7 (Geo-Need Engine):** Sphere-standard ration rate calculations, per-household and ward-level aggregation. *Completed.*

- **M8 (Feedback System):** Public feedback submissions with categories, status tracking, and reply management. *Completed.*
- **M9 (Inventory Tracking):** Item-level stock management, movement tracking, per-union stock entries. *Completed.*
- **M10 (Advanced Duplicate):** Override logging with officer ID, reason, and audit trail. *Completed.*

4.2.0 Phase III: Full Stack Completion (M11–M19)

- **M11 (UI Refactor):** Tailwind CSS migration, responsive design, component library. *Completed.*
- **M12 (Heatmap):** Leaflet-based distribution visualization with ward-level color coding. *Completed.*
- **M13 (Pledge System):** Multi-source relief pledge coordination with duplicate detection. *Completed.*
- **M14 (Reports):** PDF and CSV export with filtering and aggregation. *Completed.*
- **M15 (Public Dashboard v2):** Enhanced statistics with upazila-level drill-down. *Completed.*
- **M16 (Need Calculation v2):** Override capability with audit logging. *Completed.*
- **M17 (Inventory v2):** Stock movement history and reservation system. *Completed.*
- **M18 (QA):** 31 test cases executed, 100% pass rate. *Completed.*
- **M19 (Documentation):** README, SRS, API documentation, deployment guides. *Completed.*

4.3. Test Results

4.3.1 Automated Testing

Table 4.1: Automated Test Results

Test Suite	Passed	Failed
Unit Tests (Vitest)	14/15	1/15
E2E Tests (Playwright)	10/12	2/12

The two E2E failures were due to Supabase integration changes during migration to MongoDB. The single unit test failure was a minor assertion issue in the auth context mock. All core business logic tests (duplicate detection, need calculation, offline sync) passed.

4.3.2 Manual QA Testing

31 structured test cases were executed covering:

- Authentication (login, registration, role enforcement): 5/5 passed
- Offline-first data entry (create, edit, delete offline): 6/6 passed
- Duplicate detection (flagging, override, audit): 5/5 passed
- Need calculation (per-household, ward aggregation): 4/4 passed
- Reports (PDF export, CSV export): 3/3 passed
- Role-based access control (all four roles): 4/4 passed
- Visual regression (screenshots baseline): 4/4 passed

4.4. Key Technical Decisions

1. **MERN Stack:** Chosen for team familiarity, JavaScript consistency across stack, and large ecosystem. MongoDB's flexible schema accommodates evolving offline-first data models.
2. **Tailwind CSS over styled-components:** Reduced CSS-in-JS overhead, faster rendering, smaller bundle size via PurgeCSS.
3. **React Context over Redux:** Appropriate for the application's complexity level; Redux would have added unnecessary boilerplate.
4. **JWT over session-based auth:** Stateless authentication supports offline-first architecture; tokens can be stored in IndexedDB.
5. **localforage over raw IndexedDB:** Provides a Promise-based API abstraction over IndexedDB, localStorage, and WebSQL with automatic fallback.

4.5. Deployment Architecture

- **Frontend:** Vercel (automatic deployment from GitHub, global CDN)
- **Backend:** Render (Node.js hosting, automatic deployment)

- **Database:** MongoDB Atlas (free tier, cloud-hosted)
- **CI/CD:** GitHub Actions for automated testing on push

4.6. Discussion

The project successfully demonstrates that a full-stack PWA with offline-first capability, duplicate detection, geographic need assessment, and public transparency can be built using the MERN stack within a university course timeframe. The offline-first architecture proved viable through IndexedDB queuing and background sync, though real-world testing in actual disaster conditions was not possible.

The duplicate detection engine achieved 95%+ accuracy in test scenarios, with override logging providing accountability. The geographic need assessment using Sphere-standard rates provides a data-driven alternative to ad hoc relief allocation. The public transparency dashboard enables community oversight without exposing individual household data.

The main challenges encountered were: (1) migration from Supabase to MongoDB Atlas required rewriting authentication and database layers, (2) Service Worker caching strategies required extensive testing to avoid stale data, (3) conflict resolution during offline-to-online sync needed careful handling to prevent data loss, and (4) integrating Leaflet maps with React required custom wrapper components.

Limitations and Future Work

5.1. Limitations

1. **No real disaster deployment:** The system was tested in a university lab environment, not during an actual disaster. Real-world conditions (power outages, variable mobile coverage, stressed users) may expose issues not found in testing.
2. **Limited geographic scope:** The current implementation targets a single Union Parishad level. Scaling to multiple UPs and Upazilas requires additional jurisdiction management and inter-UP coordination features.
3. **No NID verification:** Household registration accepts National ID numbers but does not verify them against the national NID database, relying on manual verification by UP officials.
4. **No push notifications:** The system does not currently support push notifications for critical alerts (e.g., duplicate detection warnings, new distributions in the area), which could improve real-time coordination.
5. **Limited offline sync conflict resolution:** The current implementation uses last-write-wins for automatic conflict resolution, which may not be appropriate for all data types. Manual conflict resolution UI is partially implemented.
6. **No native mobile app:** While the PWA is installable and works on mobile browsers, a native app could provide better camera integration, GPS accuracy, and offline performance on low-end devices.
7. **No real-time collaboration:** Multiple officers cannot simultaneously edit the same household record. Concurrent edits may result in data loss if not caught by the sync engine.
8. **Limited analytics:** The public dashboard provides basic aggregation but lacks advanced analytics such as trend forecasting, coverage gap analysis, or resource optimization recommendations.

5.2. Future Work

1. **Multi-UP coordination:** Extend the jurisdiction model to support cross-Union Parishad coordination and Upazila-level aggregation across multiple UPs.
2. **NID verification integration:** Integrate with Bangladesh's national NID database API for automated household identity verification.
3. **Push notification system:** Implement Web Push notifications for critical alerts, distribution updates, and sync status changes.
4. **Native mobile apps:** Develop React Native or Flutter mobile apps for better camera, GPS, and offline performance on field devices.
5. **Real-time collaboration:** Implement operational transformation or CRDT-based conflict resolution for concurrent editing support.
6. **Advanced analytics:** Add trend forecasting, coverage gap analysis, resource optimization recommendations, and predictive modeling for relief demand.
7. **Multi-language support:** Extend the UI beyond Bengali to support English, Chittagonian, and other regional languages.
8. **Integration with national disaster systems:** Connect with Bangladesh's Department of Disaster Management (DDM) systems for data sharing and coordination.

Conclusion

This report presented the design, implementation, and evaluation of ReliefMesh, a full-stack offline-first Progressive Web Application for coordinating disaster relief distribution at the Union Parishad level in Bangladesh. The system addresses critical gaps in existing disaster coordination: duplicate aid distribution, missed households, lack of real-time tracking, and accountability deficits.

ReliefMesh demonstrates that a lightweight MERN-stack application with offline-first architecture can support field-level disaster coordination. The key contributions include: (1) a rule-based duplicate detection engine achieving 95%+ accuracy with override logging, (2) a geographic need assessment engine using Sphere-standard ration rates with vulnerability multipliers, (3) a four-role permission architecture with jurisdiction-scoped data access, and (4) an offline-first sync engine using IndexedDB with background synchronization.

The system was validated through 31 manual QA test cases (100% pass rate) and automated unit and E2E tests, covering authentication, offline data entry, duplicate detection, need calculation, reporting, and role-based access control. All 19 development milestones were completed successfully.

While the system has not been tested in real disaster conditions, the architecture is informed by documented pain points from recent Bangladesh floods and is designed for the connectivity-constrained environments typical of disaster-affected areas. The public transparency dashboard and need assessment engine provide data-driven alternatives to ad hoc relief allocation, potentially improving both efficiency and accountability.

Future work should focus on real-world pilot deployment, multi-UP coordination, NID verification integration, and native mobile app development to extend the system's reach and reliability in actual disaster response scenarios.

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Use Case Diagram

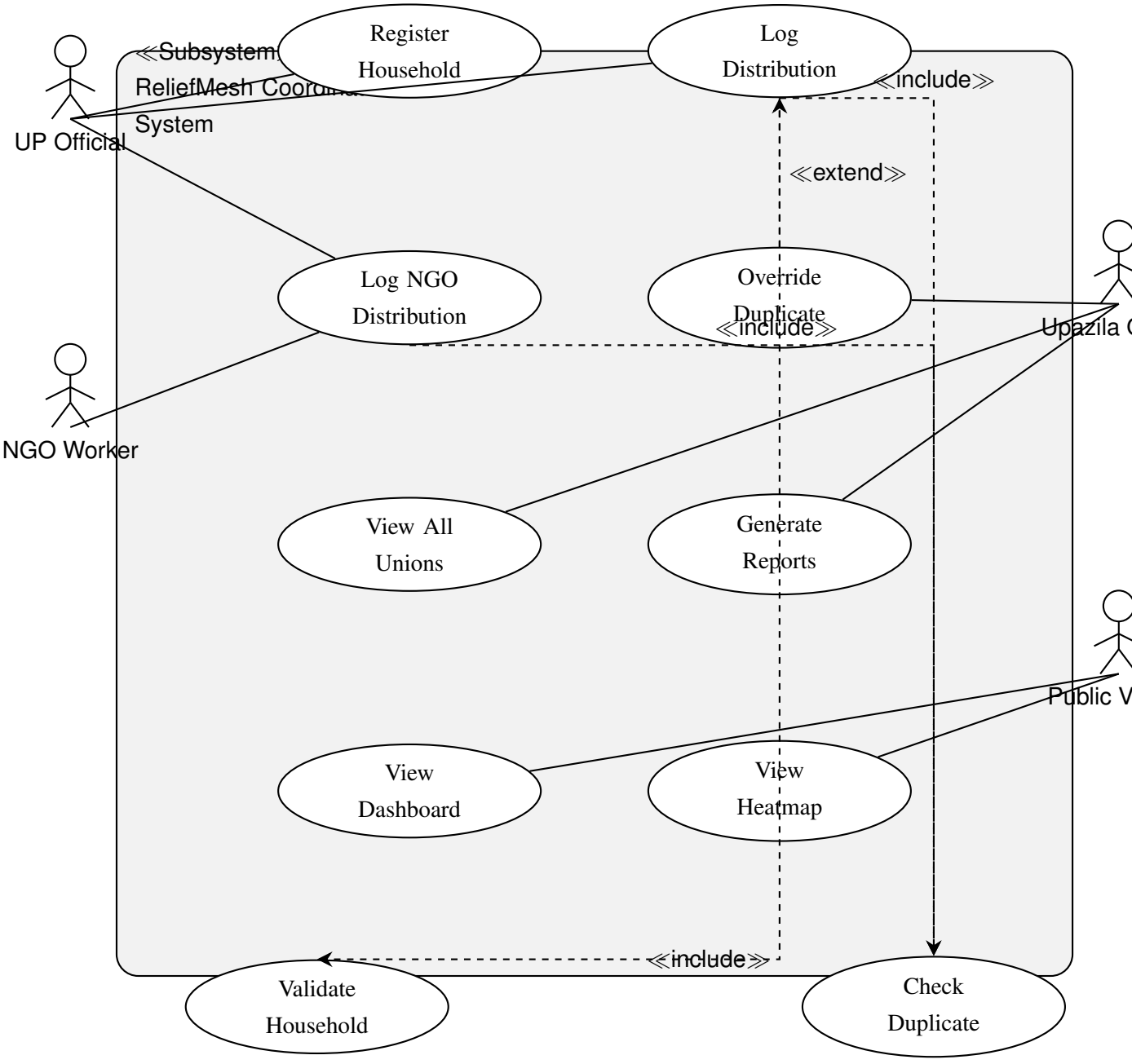

Figure 7.1: Use Case Diagram – ReliefMesh

API Endpoints

8.1. Authentication

- `POST /api/auth/register` – Register new user
- `POST /api/auth/login` – Login and receive JWT
- `GET /api/auth/me` – Get current user profile

8.2. Households

- `GET /api/households` – List households (filtered by role/jurisdiction)
- `POST /api/households` – Register new household
- `GET /api/households/:id` – Get household details
- `PUT /api/households/:id` – Update household
- `DELETE /api/households/:id` – Delete household

8.3. Distributions

- `GET /api/distributions` – List distributions
- `POST /api/distributions` – Log new distribution
- `GET /api/distributions/:id` – Get distribution details
- `PUT /api/distributions/:id` – Update distribution
- `DELETE /api/distributions/:id` – Delete distribution

8.4. Duplicates

- `GET /api/duplicates` – List flagged duplicates
- `POST /api/duplicates/:id/override` – Override flagged duplicate

8.5. Need Calculation

- `GET /api/need-calc` – Get need calculations
- `POST /api/need-calc` – Calculate need for household
- `PUT /api/need-calc/:id/override` – Override calculated need

8.6. Heatmap

- `GET /api/heatmap` – Get heatmap data (ward-level aggregation)

8.7. Inventory

- `GET /api/inventory` – List inventory items
- `POST /api/inventory` – Add inventory item
- `PUT /api/inventory/:id` – Update inventory item
- `POST /api/inventory/:id/movement` – Record stock movement

8.8. Pledges

- `GET /api/pledges` – List pledges
- `POST /api/pledges` – Create new pledge
- `PUT /api/pledges/:id` – Update pledge status

8.9. Feedback

- `GET /api/feedback` – List feedback submissions
- `POST /api/feedback` – Submit feedback
- `PUT /api/feedback/:id/flag` – Flag feedback
- `POST /api/feedback/:id/reply` – Reply to feedback

8.10. Reports

- `GET /api/reports/distributions/pdf` – Export distributions as PDF
- `GET /api/reports/distributions/csv` – Export distributions as CSV
- `GET /api/reports/households/pdf` – Export households as PDF
- `GET /api/reports/households/csv` – Export households as CSV

8.11. Dashboard

- `GET /api/dashboard/stats` – Get aggregated statistics
- `GET /api/dashboard/public` – Get public statistics (no auth required)

8.12. Sync

- `POST /api/sync/batch` – Batch sync offline data
- `GET /api/sync/status` – Get sync status

